npj | computational materials

Article

Published in partnership with the Shanghai Institute of Ceramics of the Chinese Academy of Sciences

https://doi.org/10.1038/s41524-025-01916-8

A non-orthogonal representation for materials based on chemical similarity

Check for updates

Tiago F. T. Cerqueira[1], Haichen Wang[2], Silvana Botti[3] ✉ & Miguel A. L. Marques[2] ✉

We present a novel approach to generate a fingerprint for crystalline materials that balances efficiency for machine processing and human interpretability, allowing its application in both machine learning inference and understanding of structure-property relationships. Our proposed material encoding has two components: one representing the crystal structure and the other characterizing the chemical composition, which we call Pettifor embedding. For the latter, we construct a non-orthogonal space where each axis represents a chemical element and where the angle between the axes quantifies a measure of the similarity between them. The chemical composition is then defined by the point on the unit sphere in this non-orthogonal space. We show that the Pettifor embeddings systematically outperform other commonly used elemental embeddings in compositional machine learning models. Using the Pettifor embeddings to define a distance metric and applying dimension reduction techniques, we construct a two-dimensional global map of the space of thermodynamically stable crystalline compounds. Despite their simplicity, such maps succeed in providing a physical separation of material classes according to basic physical properties.

The last decade has seen a remarkable surge in computational materials science, largely enabled by advances in high-throughput density functional theory and machine learning techniques[1,2]. These have greatly increased our knowledge and understanding of materials and contributed to the discovery of many compounds with improved properties. Inorganic materials databases catalog most experimentally verified compounds to date and millions of other hypothetical phases[3–7]. They also combine structural information with data on thermodynamic stability and a wealth of other mechanical, electronic, magnetic, etc. properties.

This explosion of data has brought with it new challenges. One particular challenge we address here is the digital representation of a compound that is both humanly understandable and suitable for material informatics techniques. In particular, these representations should allow the visualization of material properties across different compositions and structural types, greatly extending the concept of structure maps[8,9]. These were originally developed to find correlations between the structural type of a compound and the electron configuration of its constituents, providing insight into the structure of new or hypothetical compounds.

For binary compounds (or multinary compounds where only two chemical species vary) it is often easy to produce two-dimensional maps where the $x-$ and $y-$ axes run through the periodic table (see, e.g., Refs. 10–13). The order of the elements may simply reflect the atomic number, or some other ordering such as electronegativity or the Pettifor scale and its generalizations[9,12,14–16]. The latter is particularly interesting because it reflects the similarity between chemical elements and gives the systematic variation of the property over the binary composition range. Unfortunately, for ternary or multinary compounds, or when the dataset contains materials with different crystal structures, the production of such material maps is much more complicated.

In fact, it is very difficult to represent material space in a way that allows visualization—and interpretation—of how material properties vary across a given set of compounds. A common solution is to use machine learning embeddings, which are readily available from most neural network architectures. For example, in crystal graph networks[17] we can use the feature vector obtained after pooling the graph. These can then be used in conjunction with dimension reduction techniques[18,19] to produce two-dimensional (or higher dimensional) maps that can be used to visualize material properties over material space[20,21]. Unfortunately, trained embeddings already contain information about the target property (or properties), which often complicates the interpretation of structure-property relationships.

[1]CFisUC, Department of Physics, University of Coimbra, Rua Larga, 3004-516 Coimbra, Portugal. [2]Research Center Future Energy Materials and Systems of the Research Alliance Ruhr and Interdisciplinary Centre for Advanced Materials Simulation, Ruhr University Bochum, Universitätsstraße 150, D-44801 Bochum, Germany. [3]Research Center Future Energy Materials and Systems of the Research Alliance Ruhr and Interdisciplinary Center for Advanced Materials Simulations, Faculty of Physics and Astronomy, Ruhr University Bochum, Universitätsstraße 150, D-44801 Bochum, Germany. ✉e-mail: silvana.botti@rub.de; miguel.marques@rub.de

To solve this problem, we need a representation of a material in a vector space (also sometimes called fingerprint, descriptor, or embedding) based only on its chemical composition and crystal structure, such that the distances between points (i.e. compounds) reflect the degree of similarity between these compounds. We divide the fingerprint into two parts. The first should be a representation of the crystal structure of the materials. Fortunately, there are already several structural fingerprints available in the literature. We will use the one available in PYMATGEN[22], which measures the similarity between two structures based on local coordination information from all sites in the crystal structures[23]. This representation meets our twin requirements of being machine-friendly and based on a solid human understanding of the underlying physics.

For representing chemical elements, the simplest approach is one-hot encoding[24] (see Fig. 1), where each dimension corresponds to a single chemical element. More sophisticated alternatives include: (i) encoding domain knowledge through physically motivated elemental features[25,26], (ii) learning compressed elemental representations from physical properties[17,27], or (iii) extracting embeddings from materials databases[28–30] or scientific literature[31–33]. While the latter two categories consistently demonstrate superior performance in predictive tasks[34], they sacrifice interpretability, functioning as black boxes that obscure the chemical reasoning underlying predictions. We therefore introduce a new representation scheme that bridges this gap, combining the computational efficiency of learned embeddings with the transparency and physical grounding essential for human interpretation and scientific insight.

We start with a measure of the similarity between chemical elements. This could be, for example, the Euclidean distance between the initial (untrained) embeddings of the chemical elements, as provided by MATMINER[26]. Instead, we introduce a different measure based on the similarity scale proposed by some of us in ref. 35, which was constructed by simple statistical arguments using the experimentally known inorganic compounds present in ICSD[36]. Using this similarity measure, we establish a non-orthogonal space where each axis corresponds to a chemical element and the angle between axes quantifies the similarity between elements. In this framework, chemical compositions are represented as points on the unit sphere of this non-orthogonal space. The details of the method to define our similarity metric and the corresponding composition embeddings, which we call Pettifor embeddings, are contained in section IV.

We remark that recent developments in representing chemical composition have explored alternatives to Euclidean distance metrics. Notable among these are approaches based on Earth Mover's Distance (EMD) for comparing compositions[37,38], which measure the minimal "cost" of transforming one composition into another, based on data-mined definitions of chemical similarity[15,39]. Our approach differs by maintaining an Euclidean distance, but in a non-orthogonal space where the angle between the axes, and therefore the distance between two compositions, is controlled by elemental similarities and emerges naturally from observed substitution patterns in experimental crystal structures. This approach captures richer information about elemental relationships than one-dimensional ordering schemes (like the modified Pettifor scale) used to define, e.g., the Element Mover's Distance (ElMD)[37], allowing us to distinguish between neighbors of the same order of a given element by their absolute distance. In our Pettifor embeddings, the non-orthogonal geometry arises from materials co-occurrence statistics rather than optimization for specific prediction tasks. This offers two key advantages: (i) the correlations are unsupervised and independent of target properties, and (ii) the geometric structure remains directly interpretable and validatable against chemical knowledge. However, spurious correlations may arise due to bias and incompleteness in the ICSD database underlying these statistics. Several mitigation techniques have been discussed in the original publication[15], and the statistics and corresponding embeddings should be updated periodically as new experimental data become available.

Having defined the problem and outlined our approach, we now present results showing how our non-orthogonal representation captures meaningful chemical relationships, generates interpretable visualizations of material space, and provides powerful embeddings for machine learning applications, all while maintaining human-understandable connections to underlying chemical principles.

Results

Similarity of chemical elements

To demonstrate the effectiveness of our approach, we first computed the compositional fingerprints of the chemical elements, represented by the rows of the similarity matrix $\mathcal{S}$, as detailed in section IV.

We can represent the periodic table by performing a dimension reduction of our compositional embeddings, using the Uniform Manifold

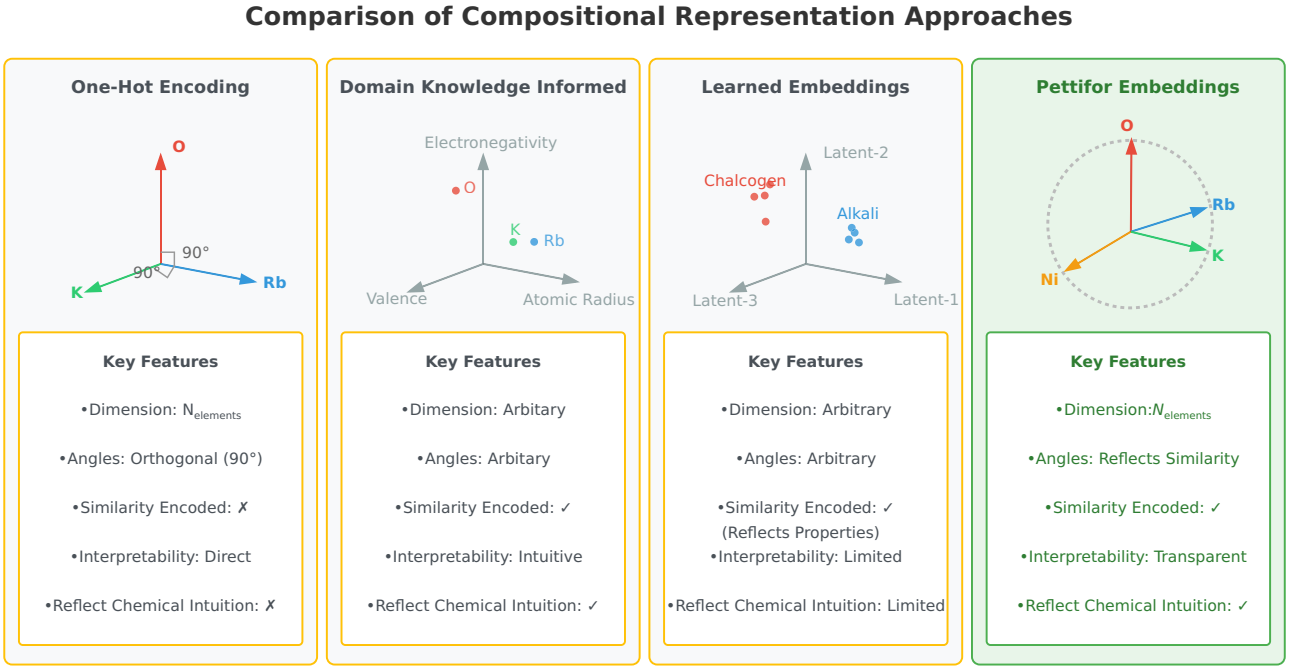


Fig. 1 | Comparison of different methods for embedding chemical elements.

Approximation and Projection (UMAP)[19]. The resulting two-dimensional map, which can be thought of as a data-mined and machine-learned periodic table, is shown in Fig. 2. We have removed noble gases (which rarely form compounds) and have colored the points according to the group of the periodic table.

From the construction, we expect similar chemical elements to be close together in the map, although the reduction to only two dimensions may distort the distances in the > 80-dimensional composition space. We also note that the actual shape of the map depends on the parameters of UMAP (such as the number of neighbors or the minimum distance used by UMAP to decide how closely points are packed together), but the relative distribution of chemical elements is quite robust. In the ESI it is possible to see the impact of different UMAP parameters on the clustering patterns and spatial distribution of chemical elements in the reduced feature space.

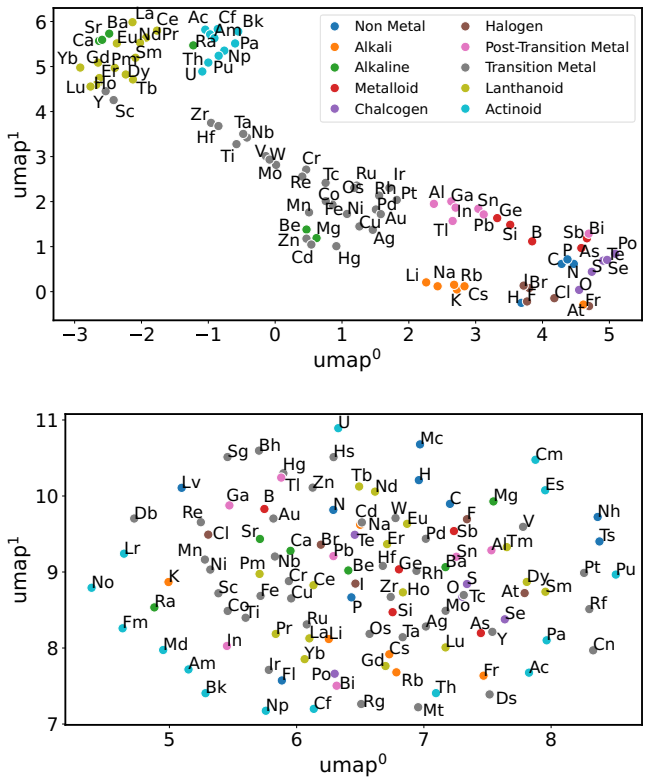


Fig. 2 | Two-dimensional maps of the chemical elements obtained by reducing the dimensions of the Pettifor embeddings (top panel) and of the one-hot embeddings (bottom panel). The axes indicate the two dimensions returned by UMAP. The points are colored in both panels according to the group of the periodic table to which the elements belong. Comparison with other common embeddings is available in the ESI.

The top panel of Fig. 2 depicts the two-dimensional map of chemical elements obtained by reducing the dimensions of the Pettifor embeddings. This visualization clearly shows chemical elements with similar properties clustered together. The lanthanides and actinides appear in the top left region, transition metals occupy the middle area of the plot, while nonmetals and alkali elements are predominantly positioned in the bottom right. This natural clustering emerges solely from the substitution patterns observed in crystal structures, without explicitly encoding traditional periodic table relationships.

The bottom panel of Fig. 2 presents, for comparison, a similar two-dimensional projection obtained using the one-hot[40] encoding representation. Unlike our Pettifor embedding, which captures chemical similarities through angles between element vectors, the one-hot representation treats all elements as equidistant from each other in the original space. Other commonly used elemental embeddings, like Magpie[41], Mat2Vec[31], Jarvis[7], and CGCNN[17], display an intermediate performance in comparison to the Pettifor and the one-hot embeddings, with Magpie achieving the second-best representation. The corresponding two-dimensional maps are shown in Fig. S1 of the ESI.

To further evaluate the effectiveness of our compositional embedding against existing approaches, we compared the elemental similarities (e.g., distances in the embedding space) captured by different methods. Table 1 shows the top three most similar elements to S, Si, and Ag according to various metrics: ElMD[37], the composition-only EMD used for the grouped representation of interatomic distances (GRID) of ref. 38, here called for simplicity C-GRID, and Euclidean distances obtained from Mat2Vec[31], Magpie[41], and our Pettifor embedding.

The metric derived from our elemental embeddings shows general agreement with ElMD, which is expected as both methods derive from the same chemical similarity matrix that was used to define the modified Pettifor scale[15]. However, significant differences emerge when compared with property-based embeddings like Magpie or text-mining approaches like Mat2Vec. For example, while our method identifies that selenium (Se) is chemically most similar to sulfur (S) with a distance of 0.77, Magpie unexpectedly suggests gallium (Ga) as most similar. The key advantage of our approach with respect to ElMD is its ability to capture nuanced differences in similarity. For instance, our embedding distinguishes between the distances of S to Se (0.77) and S to O (1.26), even if both Se and O are nearest neighbors of S in the 1D modified Pettifor scale, reflecting the much higher replaceability of S by Se than by O in real compounds. Furthermore, unlike embeddings derived from trained models, our representation remains independent of specific property prediction tasks, offering a more general framework for understanding chemical similarity.

Fingerprint for chemical composition

As a first example of application of the Pettifor embeddings to crystalline materials, we show in Fig. 3 two-dimensional maps of the dataset of perovskites with composition ABC_3 from Ref. 42. We note that the

Table 1 | Top three most similar elements to S, Si, and Ag based on ElMD[37] and C-GRID[38], as well as Euclidean distances using Mat2Vec[31], Magpie[41], CGCNN[17], and our Pettifor embeddings

Metrics	S			Si			Ag		
ElMD	Se (1)	O (1)	Te (2)	Ge (1)	B (1)	Sn (2)	Cu (1)	Au (1)	Pd (2)
C-GRID	Se (0.08)	Cr (0.10)	Be (0.12)	Ge (0.06)	As (0.17)	V (0.18)	Cu (0.07)	Au (0.28)	Li (0.30)
Mat2Vec	N (3.0)	O (3.3)	C (3.4)	Ge (2.3)	Al (2.4)	Fe (2.9)	Au (2.2)	Cu (2.5)	Pd (3.1)
Magpie	Ga (98.5)	P (98.6)	I (110.2)	Ni (59.4)	Co (98.1)	Fe (132.2)	Ge (54.2)	Pr (98.1)	La (105.4)
CGCNN	Sb (1.73)	Lu (2.0)	Yb (2.24)	In (1.73)	Yb (2.24)	Hf (2.24)	Ce (1.0)	Yb (1.73)	Hf (1.73)
Pettifor	Se (0.77)	Po (0.88)	Te (0.94)	Ge (0.64)	Sn (0.96)	Ga (1.04)	Au (0.74)	Cu (0.85)	Pd (0.90)

Distances are shown in parentheses. Note that the absolute values of distances can not be compared across different representations.

compounds in this dataset all share the same structure, namely the cubic $Pm\bar{3}m$ (space group #221), where the 1*a*, 1*b*, and 3*c* Wyckoff positions are respectively occupied by A, B, and C atoms, giving a primitive cell containing 5 atoms. The figure provides therefore only a map based on the chemical composition. As both compounds ABC_3 and BAC_3 have the same composition, and therefore the same Pettifor embeddings, we plot only the compound with the lowest energy. The points are colored according to their distance from the convex hull of thermodynamic stability, showing stability trends within and across these perovskite subfamilies. Since the chemical composition is dominated by the atom C, the map is naturally divided into clusters, one for each C. The distribution of the clusters is then largely determined by the similarity between the C atoms. Within each cluster, there is a fine structure that reflects the similarity between the A and B atoms.

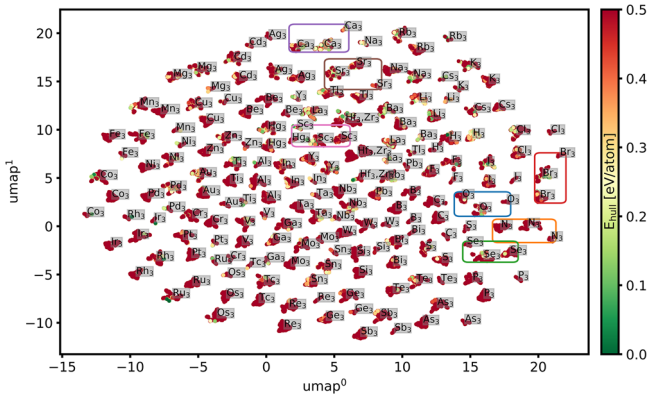


Fig. 3 | Two-dimensional composition map of cubic ABC_3 perovskites obtained by reducing the dimensions of the Pettifor embeddings with UMAP. For the plot we chose the fixed structure ABC_3 or BAC_3 with the lowest energy. Points are colored according to the distance to the convex hull of the corresponding material, capped at 0.5 eV/atom. The labels denote the C chemical element. Islands with C = {O, Br, N, Se, Ca, Sr, Sc} are highlighted.

In Fig. 3 the regions containing AB{O, N, Se, Br}$_3$ perovskites are indicated by colored boxes, showing distinct clustering patterns that are present within each perovskite family. For each cluster, we observe that compositions including transition metals form the main clusters, while other cations form smaller islands. It is clear from the color code of Fig. 3 that most cubic perovskites are highly unstable, but some clusters of higher stability can be seen. The well-known oxide perovskites ABO_3 are centered at the coordinates (15.3, 2.5). However, most stable (or near stable) compounds are inverted perovskites with an H, C, N, O, etc. in the Wyckoff 1b position. These can be seen as green dots usually at the boundaries of the clusters.

By far the most stable systems are the inverted perovskites of the type $ABCa_3$ around (3.8, 18.8), $ABSr_3$ around (5.2, 15.9), $ABSc_3$ around (3.2, 8.8), etc.

Table 2 | Mean absolute error across various datasets[34] comparing different feature representations for CrabNet: MAT2VEC, MAGPIE, CGCNN, one-hot encoding[40], and our proposed non-orthogonal representation

Dataset	Set size	MAT2VEC	MAGPIE	CGCNN	ONE-HOT	This work
steels_yield	312	128	207	132	**126**	136
jdft2d	636	41.7	40.0	**37.8**	41.4	40.8
phonons	1265	78.3	134	**61.4**	82.3	73.2
Aflow thermal expansion	3421	4.70×10^{-6}	10.7×10^{-6}	$\mathbf{4.17 \times 10^{-6}}$	4.80×10^{-6}	4.24×10^{-6}
Aflow thermal cond.	3422	2.50	3.30	**2.29**	2.48	2.31
Aflow bulk modulus	3428	9.89	21.4	9.95	9.97	**9.34**
Aflow Debye temp.	3428	36.6	56.8	**33.7**	37.0	34.2
Aflow shear modulus	3428	10.0	15.4	9.62	10.0	**9.48**
MP shear modulus	4328	13.2	15.9	12.4	13.2	**12.2**
MP bulk modulus	4414	12.5	21.8	11.7	12.6	**11.2**
MP elastic anisotropy	4431	8.29	**8.16**	8.16	8.16	8.23
Exp E_{gap}	4604	0.368	0.576	**0.335**	0.362	0.352
dielectric	4764	0.271	0.379	0.254	0.269	**0.254**
Exp is metal (%)	4921	95.6	93.2	96.6	95.7	**96.7**
glass (%)	5680	90.4	69.9	**91.1**	90.7	90.1
$\log_{10}(G_{VRH})$	10987	0.105	0.138	0.0993	0.105	**0.0951**
$\log_{10}(K_{VRH})$	10987	0.0777	0.111	0.0756	0.0782	**0.0723**
Aflow E_{gap}	19330	0.331	0.451	0.321	0.330	**0.309**
Aflow energy/atom	19346	**0.109**	0.342	0.117	0.112	0.110
MP Ehull	39663	0.0983	0.127	0.0960	0.0983	**0.0922**
MP μ_b	39663	2.29	3.79	2.50	2.31	**2.22**
CritExam Ed	59509	0.0651	0.0843	0.0630	0.0651	**0.0608**
CritExam Ef	59509	**0.0765**	0.188	0.0821	0.0773	0.0800
OQMD bandgap	239125	0.0592	0.107	0.0593	**0.0580**	0.0588
OQMD energy/atom	239190	**0.0527**	0.0696	0.0901	0.0525	0.0530
OQMD form. Enthalpy	239190	**0.0423**	0.0606	0.0530	0.0421	0.0426
OQMD volume/atom	239190	**0.329**	0.421	0.405	0.329	0.330

Note that for the classification tasks 'Exp is metal' and 'glass', the metric shown is Area Under the Receiver Operating Characteristic Curve (ROC AUC), where higher values indicate better performance. Detailed information on the benchmark datasets can be found in ref. 34.

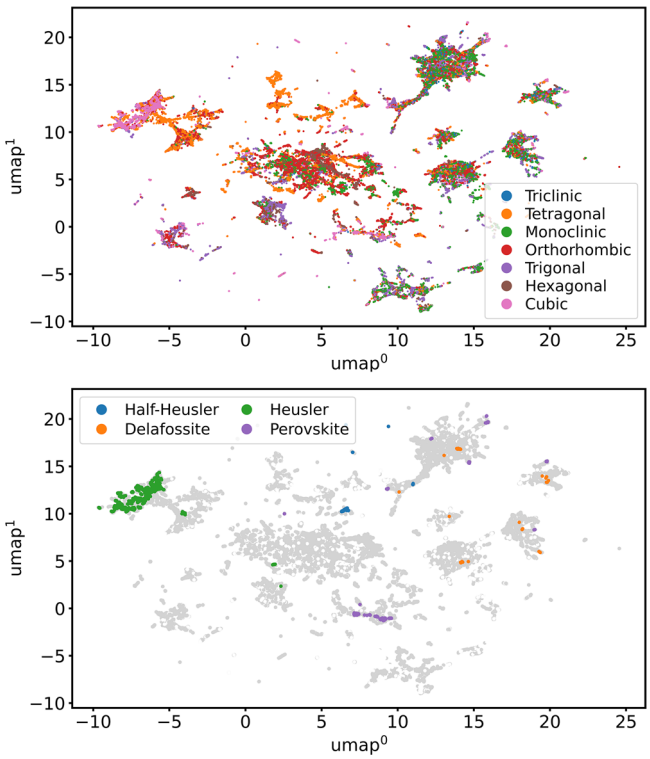


Fig. 4 | Materials on the convex hull of Alexandria, projected into two dimensions using UMAP on our Pettifor ⊕ STR embeddings, taking into account both crystal structure and chemical composition. (Top) We distinguish by crystal system. (Bottom) We highlight some well-known materials families.

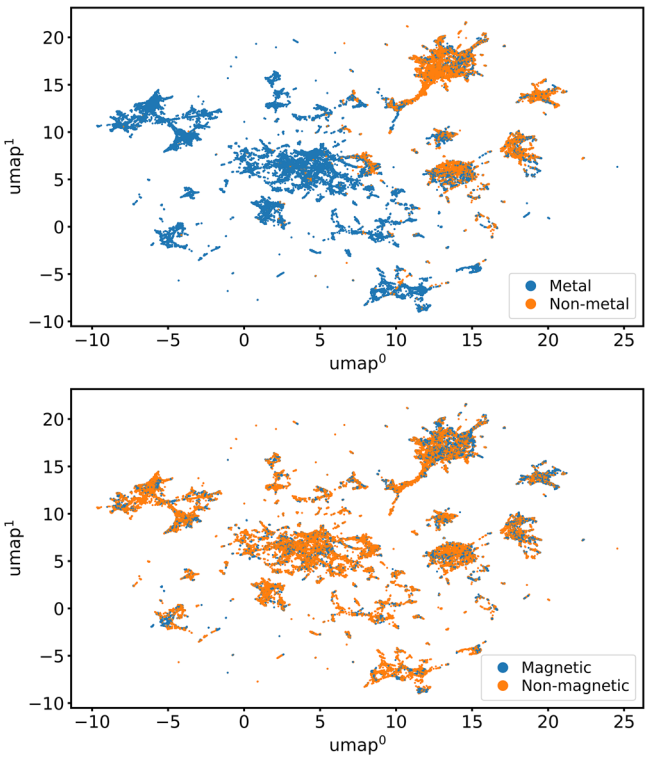


Fig. 5 | Materials on the convex hull of Alexandria, projected into two dimensions using UMAP with our Pettifor ⊕ STR embeddings. We distinguish between metallic and non-metallic (top), as well as magnetic and non-magnetic materials (bottom).

We further benchmark the performance of our Pettifor embeddings to represent compounds in compositional neural network models. Specifically, we use the compositionally restricted attention-based network (CrabNet[34]), trained with different embeddings (MAT2VEC[31], MAGPIE[41], CGCNN[17], the one-hot representation, and Pettifor) for predicting a variety of materials properties. We present in Table 2 results for 27 benchmarks[34] covering a wide range of material properties and dataset sizes. We note that for the CGCNN embeddings, we used the initial atomic features and not the trained embeddings, which would change depending on the target training property. The results are averaged over 8 different runs in order to decrease the variability due to the training. We see from the table that our constructed representation yields on average the best results while retaining simplicity and interpretability. Moreover, it performs particularly well for small- and medium-sized datasets. This improved performance suggests that our representation effectively captures the essential chemical relationships relevant to materials properties, making it especially valuable for materials discovery problems where extensive training data may not be available. Of course, our fingerprint can also be used as a node embedding in graph neural networks (or in other architectures), but this is beyond the scope of this work.

Embeddings with structural information

To compute structure maps of compounds with arbitrary structures, we can concatenate the composition and structure fingerprints, obtaining the Pettifor ⊕ STR embeddings. The two components can be combined with different weights, depending on whether one wants to give more importance to the composition or the structure. For simplicity, we have chosen equal weights for both descriptors in the following.

As a first example of application of the Pettifor ⊕ STR embeddings, we plot in Fig. 4 the map of all thermodynamically stable materials, i.e. compounds that lie on the convex hull of thermodynamic stability, found in the ALEXANDRIA[3] database. This corresponds to over 115 thousand compounds with a wide variety of chemical compositions and structural types. To reduce the number of dimensions to two, we again use UMAP. We color the points according to the property that we want to highlight, e.g. the crystal system (Fig. 4), the existence of an electronic band gap (top panel of Fig. 5), or the existence of finite magnetic moments (bottom panel of Fig. 5).

As a full discussion of the materials on the convex hull and their properties goes well beyond the scope of this work, we limit ourselves to a few general observations. UMAP divides the vast majority of the compounds into large “continents”, with a few materials scattered in smaller islands. While the actual two-dimensional map is dependent on the parameters of UMAP, it is reasonable to interpret these smaller islands as more structurally and compositionally exotic compounds with few related materials on the convex hull. It is interesting to see in Fig. 4 that, despite the extremely large diversity of the data set, the map provides a reasonable structural separation of compounds — indeed, some of the continents and islands are dominated by compounds of a particular crystal system. Furthermore, other islands correctly identify similar structural motifs, such as the hexagonal and trigonal symmetry adopted by many layered materials, or the cubic, tetragonal, orthorhombic sequence often obtained for many compounds as symmetry decreases with, for example, decreasing temperature.

UMAP combined with our fingerprint provides an excellent separation between metals and non-metals, as shown in the top panel of Fig. 5. However, such a good separation is not visible in the bottom panel, which represents magnetic and non-magnetic compounds. This suggests that, unfortunately, this latter distinction will be much more difficult to predict with machine learning approaches than the previous properties.

Discussion

An interesting question concerns the actual number of dimensions needed in practice to represent all the materials on the hull. Our Pettifor ⊕ STR vector contains $61 \times 4 = 244$ elements to represent the structure,

to which we add one extra dimension per chemical element. The question is how many of these > 300 dimensions are really required. We can get this information by performing a principal component analysis of the data. We find that only 7 components are sufficient to represent 60% of the variance in composition, while 6 are required to represent 60% of the variance in composition and structure. If we ask for 95%, we need 39 components for composition and 42 for composition and structure. This is much smaller than the original number of components, meaning that it may be possible to compress the information into a smaller feature vector that is more suitable for efficient machine learning of materials data.

To demonstrate the practical utility of our Pettifor ⊕ STR embeddings, we examine at last three concrete applications that illustrate how they can advance materials research and discovery.

First, we applied our embeddings to identify compounds with similar electronic properties. We consider as an example the crystalline materials that are closest to diamond silicon according to the distances in our embedding space. While Ge (distance 0.64) and SiC (distance 0.70) are predictably close to Si, our method also identifies less obvious compounds like $NiSi_3P_4$ (distance 0.71) that retain similar electronic properties. In Fig. 6 we show the band structure of the latter material. Figure S5 of the ESI displays for further comparison the band structures of the five nearest neighbors to Si. Such examples demonstrate how our embeddings effectively captures both chemical and electronic similarities between materials, potentially accelerating the discovery of functional analogues to known materials.

Second, using the conventional superconductor MgB_2 as a case study, we identified structurally and chemically related compounds with potentially similar properties. The closest compound, Mg_4AlB_{10} (distance 0.16), is a five-layer supercell of MgB_2 with one Mg layer replaced by Al, preserving similar band dispersion patterns. Interestingly, our method identified $LiMgB_4$ (which lies 0.123 eV/atom above the convex hull) as having a small distance to MgB_2, with a calculated superconducting critical temperature of approximately 20.8 K. The band structures of the five most similar systems to MgB_2 can be examined in the ESI. This example illustrates how our embeddings can uncover promising candidates with a desired functionality without performing any calculation, as the proximity in our representation space correlates with comparable physical properties.

Third, we introduce a new descriptor for materials on the convex hull: the "uniqueness" U, defined as the minimal Euclidean distance to neighboring materials in our embedding space. A high value of this metric reveals truly distinctive compounds that may merit special attention in experimental studies. We found that As is the most unique elementary substance (distance 1.06 to P), while hexagonal BN is the most unique binary compound on the convex hull (distance 1.17 to graphite). Despite being isovalent and isostructural, hexagonal BN and graphite exhibit dramatically different electronic properties: the former is a large-gap insulator while the latter is a semimetal. This measure of uniqueness offers valuable insights for both theoretical and experimental materials design by highlighting compounds with few similar alternatives.

In conclusion, we propose a fingerprint that provides a simple and interpretable representation of a compound, considering both its chemical composition and crystal structure. This is designed so that the distance between embeddings of compounds that are chemically similar —and therefore likely to have similar material properties—is smaller than for unrelated materials. We show how this fingerprint can be used to create structure maps, or more generally, property maps, spanning entire families of materials, or even the entire material space. Such maps

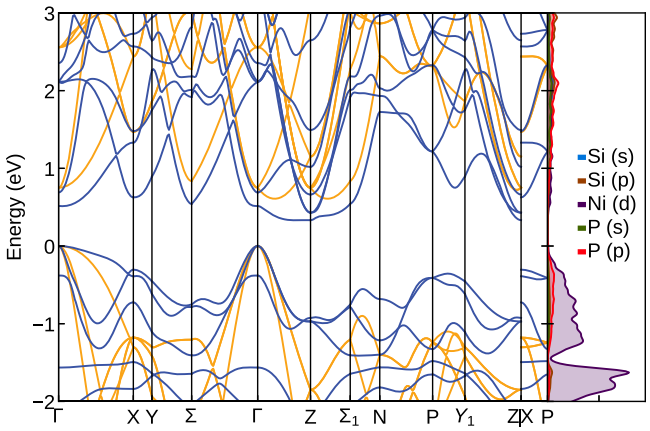


Fig. 6 | Band structures of $NiSi_3P_4$ The upfolded band structure of Si is displayed in orange lines for comparison.

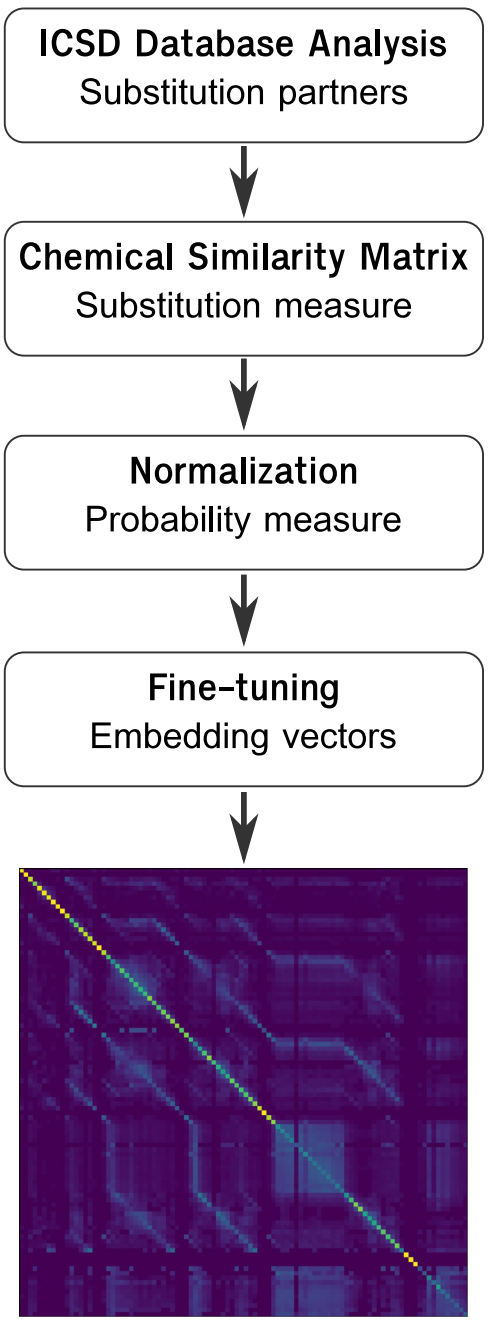


Fig. 7 | Schema depicting the construction of the matrix $\mathcal{S}$. Each line of the matrix is an unit vector that represents a chemical elements in the non-orthogonal compositional space.

can be used to visualize and interpret how material properties change across chemical space. Finally, we show that our human-generated fingerprint can compete with machine-learned opaque representations when it is used as an input feature to machine learning models. There are still some shortcomings in our approach, such as the lack of information for rare gases or some actinides, but we believe that our representation of the chemical space can already be an important tool both for accurate machine prediction of material properties and for human interpretation of high-throughput investigations.

Methods

Pettifor embeddings

We start with the raw data from Ref. 35, provided as a symmetric matrix $\mathcal{M}$ of dimension N (the total number of chemical elements). The off-diagonal elements of this matrix, $\mathcal{M}(A, B)$, record, for each pair of elements (A, B), the number of compounds in ICSD that share the same crystal structure but where A is substituted by B. The diagonal elements contain the total number N_A of ICSD compounds that include the element A, which can be interpreted as self-substitutions. Each row is then normalized to unity, yielding the (not anymore symmetric) matrix $\widetilde{\mathcal{S}}$ whose elements $\widetilde{\mathcal{S}}(A, B)$ can be interpreted as the probability of substitution $A \rightarrow B$.

To mitigate the intrinsic bias in the ICSD, where off-diagonal components are typically underestimated relative to the diagonal, we followed a procedure similar to the one already described in Ref. 15 and redistributed the weight of diagonal and off-diagonal terms by raising $\widetilde{\mathcal{S}}(A, B)$ to the power of $\alpha = 1/2$, followed by row-wise renormalization. The resulting matrix, denoted $\mathcal{S}$, defines a set of unit vectors $\mathcal{S}_A$ (the rows of $\mathcal{S}$) representing the chemical elements A in $\mathbb{R}^N$. These vectors span a non-orthogonal compositional space, where the dot product between $\mathcal{S}_A$ and $\mathcal{S}_B$ (i.e., the cosine of the angle between them) quantifies the similarity between the elements A and B. Completely dissimilar elements correspond to orthogonal unit vectors, with the angle between vectors decreasing as similarity increases. Although the dimensionality of the space remains N, the hyperpolyhedron defined by the non-orthogonal vectors occupies a smaller volume than the hypercube spanned by the orthogonal one-hot basis. A schematic representation of this construction is shown in Figs. 7 and 8.

To obtain the fingerprint of a given composition, we first construct a composition vector $\mathbf{c}$ of dimension N_{elements}, where each component c_i represents the stoichiometric fraction of element i. The fingerprint $\mathbf{f}$ is then obtained by $\mathbf{f} = \mathbf{c} \times \mathcal{S}$, yielding the Cartesian coordinates of the composition in the non-orthogonal representation.

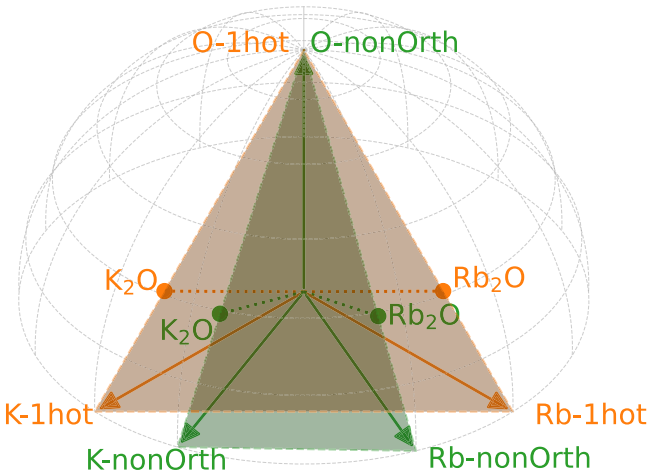


Fig. 8 | Schema depicting the Euclidean distance between K_2O and Rb_2O in a one-hot representation (orange) and in our non-orthogonal (green) space. The unit-sphere in compositional space, where the elementary substances reside (in both representations), is also indicated in gray as a guide to the eye.

Note that the vectors $\mathbf{f}$ describing chemical compositions are in general not localized on the unit sphere, but on the surface of an $(N-1)$-dimensional simplex embedded in $\mathbb{R}^N$. The reason is that they are not normalized using the Euclidean L_2 norm, but using the L_1 norm ($\sum_i c_i = 1$). The exceptions are, obviously, the elementary substances. For instance, when $N = 3$ (a ternary phase diagram), the compositional vectors lie within or on the surface (the edges) of a triangle embedded in $\mathbb{R}^3$ (see the shaded orange and green triangles in Fig. 8). While non-orthogonality determines how the $\mathcal{S}_i$ vectors deviate from the standard basis (one-hot vectors), it does not alter the Euclidean structure of $\mathbb{R}^N$. Therefore, Euclidean distances between chemical compositions remain well-defined and meaningful.

An example is shown in Fig. 8 where we plot K_2O and Rb_2O in both the standard one-hot and our non-orthogonal representations. The angle between K and Rb in our representation is about 43° (while K and Rb remain orthogonal to O due to their dissimilarity), making the distance between the two compounds closer than in a one-hot representation.

We call the composition embeddings Pettifor embeddings. By concatenating the latter with crystal structure embeddings[23] we obtain the Pettifor ⊕ STR embeddings.

Data availability

The data used in this work, along with the accompanying example notebook, is available on GitHub at https://github.com/hyllios/utils/tree/main/similarity.

Received: 16 June 2025; Accepted: 5 December 2025;
Published online: 15 January 2026

References

1. Schmidt, J., Marques, M. R. G., Botti, S. & Marques, M. A. L. Recent advances and applications of machine learning in solid-state materials science. *npj Comput. Mater.* **5**, https://doi.org/10.1038/s41524-019-0221-0 (2019).
2. Kulik, H. J. et al. Roadmap on machine learning in electronic structure. *Electron. Struct.* **4**, 023004 (2022).
3. Schmidt, J. et al. Machine-learning-assisted determination of the global zero-temperature phase diagram of materials. *Adv. Mater.* **35**, 2210788 (2023).
4. Jain, A. et al. Commentary: The materials project: A materials genome approach to accelerating materials innovation. *APL Mater.* **1**, 011002 (2013).
5. Curtarolo, S. et al. AFLOWLIB.ORG: A distributed materials properties repository from high-throughput ab initio calculations. *Comput. Mater. Sci.* **58**, 227–235 (2012).
6. Saal, J. E., Kirklin, S., Aykol, M., Meredig, B. & Wolverton, C. Materials design and discovery with high-throughput density functional theory: the open quantum materials database (oqmd). *JOM* **65**, 1501–1509 (2013).
7. Choudhary, K. et al. The joint automated repository for various integrated simulations (jarvis) for data-driven materials design. *npj Comput. Mater.* **6**, 173 (2020).
8. Villars, P., Mathis, K., Hulliger, F., De Boer, F. & Pettifor, D. *Environment Classification and Structural Stability Maps*, **1** (Elsevier Science Publishing BV, 1989).
9. Pettifor, D. G. A chemical scale for crystal-structure maps. *Solid State Commun.* **51**, 31–34 (1984).
10. Silva, A., Cao, J., Polcar, T. & Kramer, D. Pettifor maps of complex ternary two-dimensional transition metal sulfides. *npj Comput. Mater.* **8**, https://doi.org/10.1038/s41524-022-00868-7 (2022).
11. Fukuda, M., Zhang, J., Lee, Y.-T. & Ozaki, T. A structure map for AB2 type 2D materials using high-throughput dft calculations. *Mater. Adv.* **2**, 4392–4413 (2021).
12. Silva, A., Cao, J., Polcar, T. & Kramer, D. Design guidelines for two-dimensional transition metal dichalcogenide alloys. *Chem. Mater.* **34**, 10279–10290 (2022).

13. Chen, W. et al. A map of single-phase high-entropy alloys. *Nat. Commun.* **14**, https://doi.org/10.1038/s41467-023-38423-7 (2023).
14. Pettifor, D. The structures of binary compounds. i. phenomenological structure maps. *J. Phys. C: Solid State Phys.* **19**, 285 (1986).
15. Glawe, H., Sanna, A., Gross, E. & Marques, M. A. The optimal one dimensional periodic table: a modified pettifor chemical scale from data mining. *New J. Phys.* **18**, 093011 (2016).
16. Allahyari, Z. & Oganov, A. R. Nonempirical definition of the mendeleev numbers: organizing the chemical space. *J. Phys. Chem. C* **124**, 23867–23878 (2020).
17. Xie, T. & Grossman, J. C. Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Phys. Rev. Lett*. **120**, https://doi.org/10.1103/PhysRevLett.120.145301 (2018).
18. Pearson, K. Liii. on lines and planes of closest fit to systems of points in space. *London Edinburgh Philos. Mag. J. Sci.* **2**, 559–572 (1901).
19. McInnes, L., Healy, J. & Melville, J. UMAP: Uniform Manifold Approximation and Projection. *JOSS.* **3**, 861 (2018).
20. Xie, T. & Grossman, J. C. Hierarchical visualization of materials space with graph convolutional neural networks. *J. Chem. Phys*. **149**, https://doi.org/10.1063/1.5047803 (2018).
21. Wang, A. Y.-T., Mahmoud, M. S., Czasny, M. & Gurlo, A. Crabnet for explainable deep learning in materials science: Bridging the gap between academia and industry. *Integr. Mater. Manuf. Innov.* **11**, 41–56 (2022).
22. Ong, S. P. et al. Python materials genomics (pymatgen): a robust, open-source python library for materials analysis. *Comput. Mater. Sci.* **68**, 314–319 (2013).
23. Zimmermann, N. E., Horton, M. K., Jain, A. & Haranczyk, M. Assessing local structure motifs using order parameters for motif recognition, interstitial identification, and diffusion path characterization. *Front. Mater.* **4**, 34 (2017).
24. Jha, D. et al. Elemnet: Deep learning the chemistry of materials from only elemental composition. *Sci. Rep*. **8**, https://doi.org/10.1038/s41598-018-35934-y (2018).
25. Oliynyk, A. O. et al. High-throughput machine-learning-driven synthesis of full-heusler compounds. *Chem. Mater.* **28**, 7324–7331 (2016).
26. Ward, L. et al. Matminer: An open source toolkit for materials data mining. *Comput. Mater. Sci.* **152**, 60–69 (2018).
27. Herr, J. E., Koh, K., Yao, K. & Parkhill, J. Compressing physics with an autoencoder: creating an atomic species representation to improve machine learning models in the chemical sciences. *J. Chem. Phys*. **151**, https://doi.org/10.1063/1.5108803 (2019).
28. Antunes, L. M., Grau-Crespo, R. & Butler, K. T. Distributed representations of atoms and materials for machine learning. *npj Comput. Mater*. **8**, https://doi.org/10.1038/s41524-022-00729-3 (2022).
29. Jin, L. et al. Transformer-generated atomic embeddings to enhance prediction accuracy of crystal properties with machine learning. *Nat. Commun*. **16**, https://doi.org/10.1038/s41467-025-56481-x (2025).
30. Li, Y. et al. Element2Vec: Build chemical element representation from text for property prediction. Preprint at https://arxiv.org/abs/2510.13916 (2025).
31. Tshitoyan, V. et al. Unsupervised word embeddings capture latent knowledge from materials science literature. *Nature* **571**, 95–98 (2019).
32. Venugopal, V. & Olivetti, E. Matkg: an autonomously generated knowledge graph in material science. *Sci. Data* **11**, https://doi.org/10.1038/s41597-024-03039-z (2024).
33. Ye, Y. et al. Construction and application of materials knowledge graph in multidisciplinary materials science via large language model. Preprint at https://arxiv.org/abs/2404.03080 (2024).
34. Wang, A. Y.-T., Kauwe, S. K., Murdock, R. J. & Sparks, T. D. Compositionally restricted attention-based network for materials property predictions. *npj Comput. Mater*. **7**, https://doi.org/10.1038/s41524-021-00545-1 (2021).
35. Wang, H.-C., Botti, S. & Marques, M. A. Predicting stable crystalline compounds using chemical similarity. *npj Comput. Mater.* **7**, 12 (2021).
36. Zagorac, D., Müller, H., Ruehl, S., Zagorac, J. & Rehme, S. Recent developments in the inorganic crystal structure database: theoretical crystal structure data and related features. *J. Appl. Crystallogr.* **52**, 918–925 (2019).
37. Hargreaves, C. J., Dyer, M. S., Gaultois, M. W., Kurlin, V. A. & Rosseinsky, M. J. The earth mover's distance as a metric for the space of inorganic compositions. *Chem. Mater.* **32**, 10610–10620 (2020).
38. Zhang, R.-Z., Seth, S. & Cumby, J. Grouped representation of interatomic distances as a similarity measure for crystal structures. *Digit. Discov.* **2**, 81–90 (2023).
39. Hautier, G., Fischer, C., Ehrlacher, V., Jain, A. & Ceder, G. Data mined ionic substitutions for the discovery of new compounds. *Inorg. Chem.* **50**, 656–663 (2011).
40. Harris, S. & Harris, D.*Digital Design and Computer Architecture* (Morgan Kaufmann, 2015).
41. Ward, L., Agrawal, A., Choudhary, A. & Wolverton, C. A general-purpose machine learning framework for predicting properties of inorganic materials. *npj Comput. Mater*. **2**, https://doi.org/10.1038/npjcompumats.2016.28 (2016).
42. Schmidt, J. et al. Predicting the thermodynamic stability of solids combining density functional theory and machine learning. *Chem. Mater.* **29**, 5090–5103 (2017).

Acknowledgements

T.F.T.C. acknowledges financial support from Fundação para a Ciência e Tecnologia (FCT), I.P., through the project CEECINST/00152/2018/CP1570/CT0006 with DOI identifier https://doi.org/10.54499/CEECINST/00152/2018/CP1570/CT0006. S.B. acknowledges funding from the Volkswagen Stiftung (Momentum) through the project "dandelion" and from the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) through the project BO 4280/11-1.

Author contributions

T.F.T.C. and M.A.L.M. developed the Pettifor embedding method. T.F.T.C. and H.C.W. implemented the method in Python. T.F.T.C. trained the CrabNet models. T.F.T.C. and H.C.W. prepared the figures. All authors contributed to the editing and revision of the manuscript. M.A.L.M. and S.B. supervised the project and secured funding.

Funding

Open Access funding enabled and organized by Projekt DEAL.

Competing interests

Silvana Botti is an Associate Editor for npj Computational Materials, but was not involved in the editorial review of, or the decision to publish this article. All other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41524-025-01916-8.

Correspondence and requests for materials should be addressed to Silvana Botti or Miguel A. L. Marques.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

© The Author(s) 2025